

1.6 GENERAL DESCRIPTION OF THE HELICOPTER

The BO 105 is a five-place, multi-purpose, utility helicopter having a hingeless, four-bladed main rotor and a semi-rigid, two-bladed tail rotor. The main and tail rotor blades are constructed of fiber-reinforced composite materials.

Engines

The helicopter is powered by two Allison 250-C20 series engines of the freewheel turbine type. The engine power is transmitted to the main rotor transmission via independent drive systems.

Transmission

The main transmission consists of three major subassemblies in which two bevel gear input stages with freewheel clutches, a spur collector gear stage, and a planetary reduction stage are housed.

Fuel Tanks

The total fuel quantity is stored in three flexible bladder-type tanks, two main tanks (forward and aft) and one supply tank, all installed in the tank compartment beneath the cabin floor.

1.7 HELICOPTER DIMENSIONS

Figure 1-1 provides a three-view drawing of the helicopter with the respective principal exterior dimensions.

Locations on and within the helicopter can be determined in relation to fuselage stations (F.S. or STA.), waterlines (W.L.), and buttock lines (B.L.) measured in millimeters from known reference points.

1.8 CONVERSION CHARTS

See Fig. 1-2 thru 1-7.

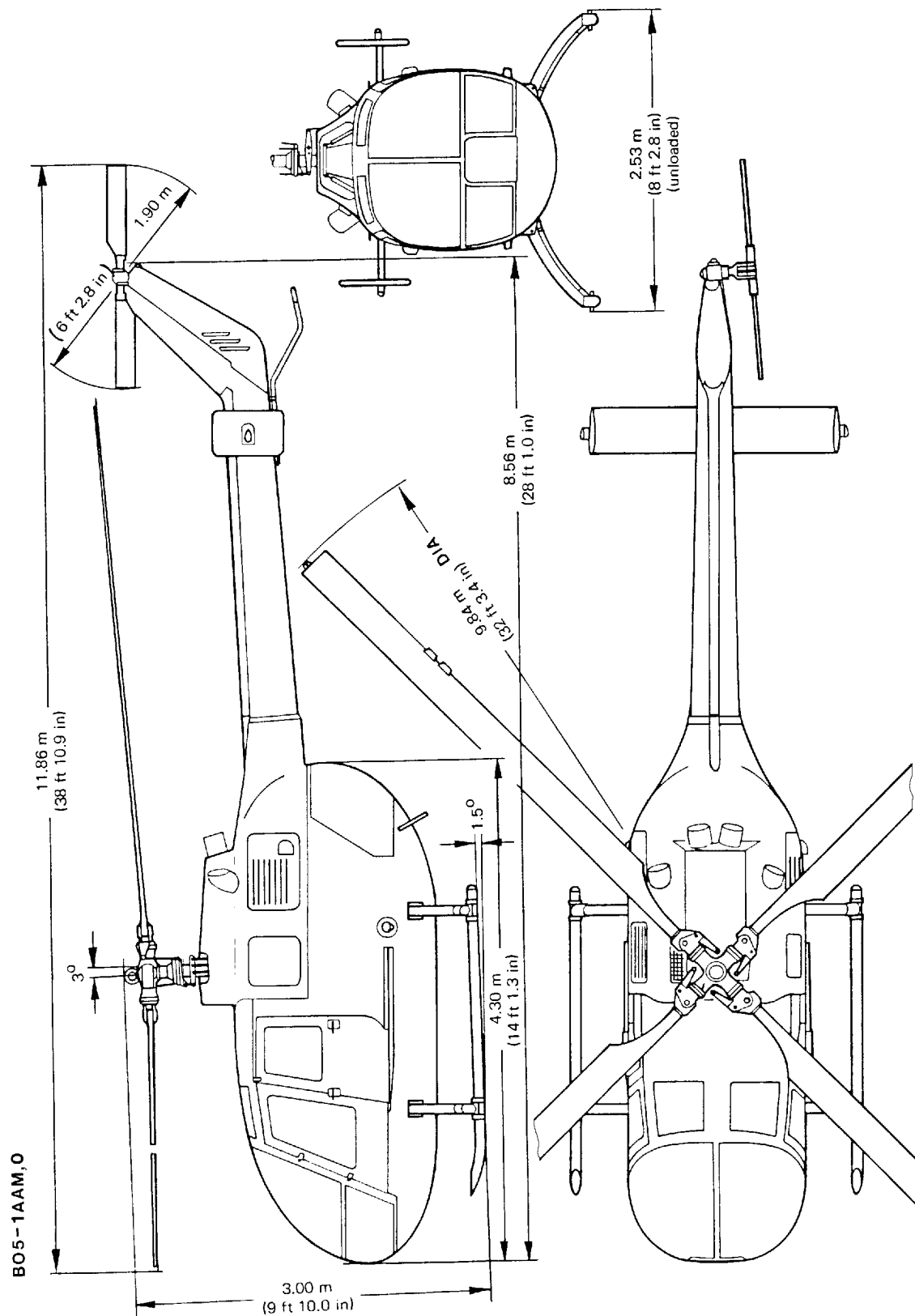


Fig. 1-1 Principal Dimensions

1.6 GENERAL DESCRIPTION OF THE HELICOPTER

The BO 105 is a five-place, multi-purpose, utility helicopter having a hingeless, four-bladed main rotor and a semi-rigid, two-bladed tail rotor. The main and tail rotor blades are constructed of fiber-reinforced composite materials.

Engines

The helicopter is powered by two Allison 250-C20 series engines of the freewheel turbine type. The engine power is transmitted to the main rotor transmission via independent drive systems.

Transmission

The main transmission consists of three major subassemblies in which two bevel gear input stages with freewheel clutches, a spur collector gear stage, and a planetary reduction stage are housed.

Fuel Tanks

The total fuel quantity is stored in three flexible bladder-type tanks, two main tanks (forward and aft) and one supply tank, all installed in the tank compartment beneath the cabin floor.

1.7 HELICOPTER DIMENSIONS

Figure 1-1 provides a three-view drawing of the helicopter with the respective principal exterior dimensions.

Locations on and within the helicopter can be determined in relation to fuselage stations (F.S. or STA.), waterlines (W.L.), and buttock lines (B.L.) measured in millimeters from known reference points.

1.8 CONVERSION CHARTS

See Fig. 1-2 thru 1-7.

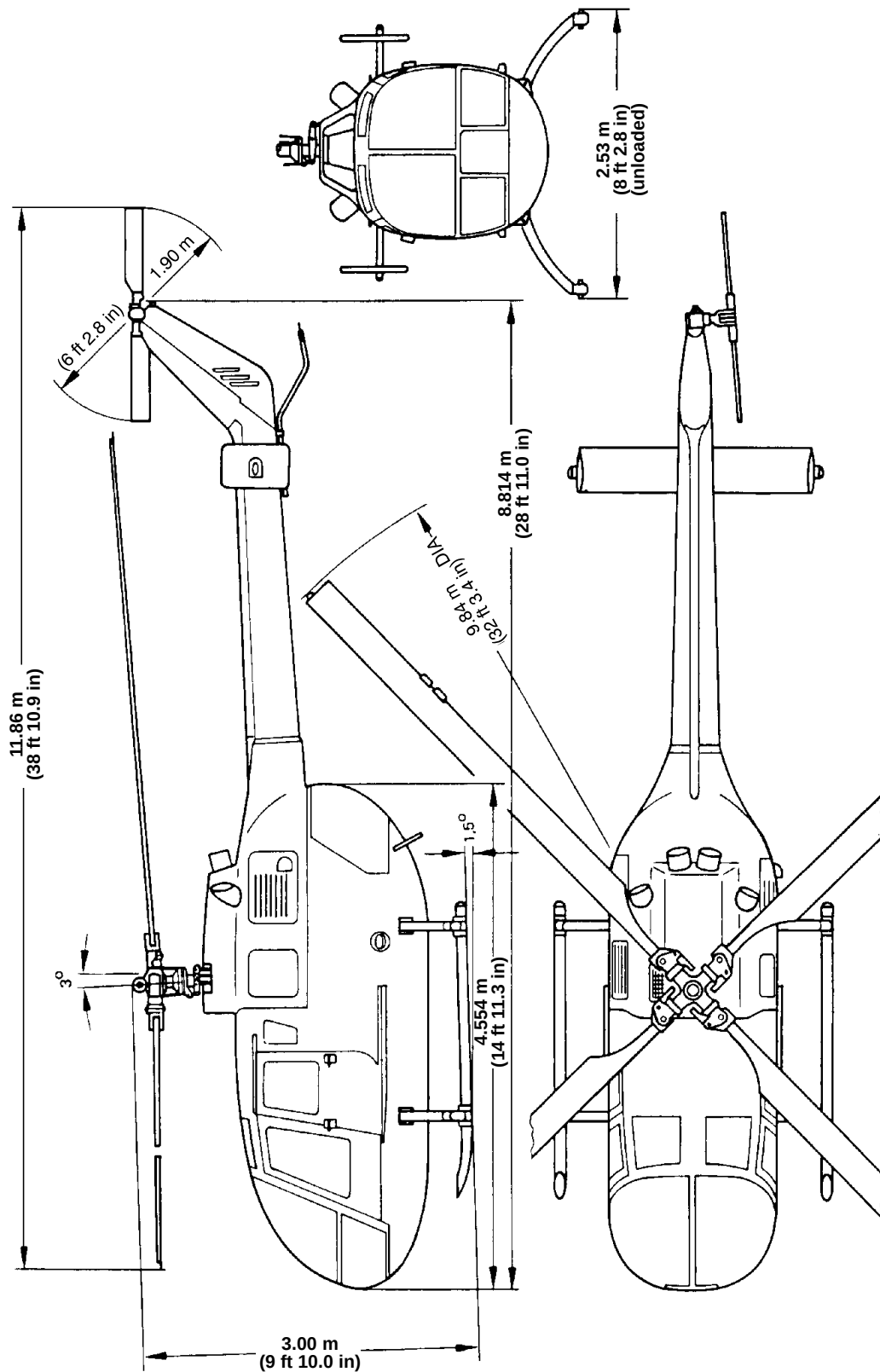


Fig. 1-1 Principal Dimensions

BO5-1ACM.0

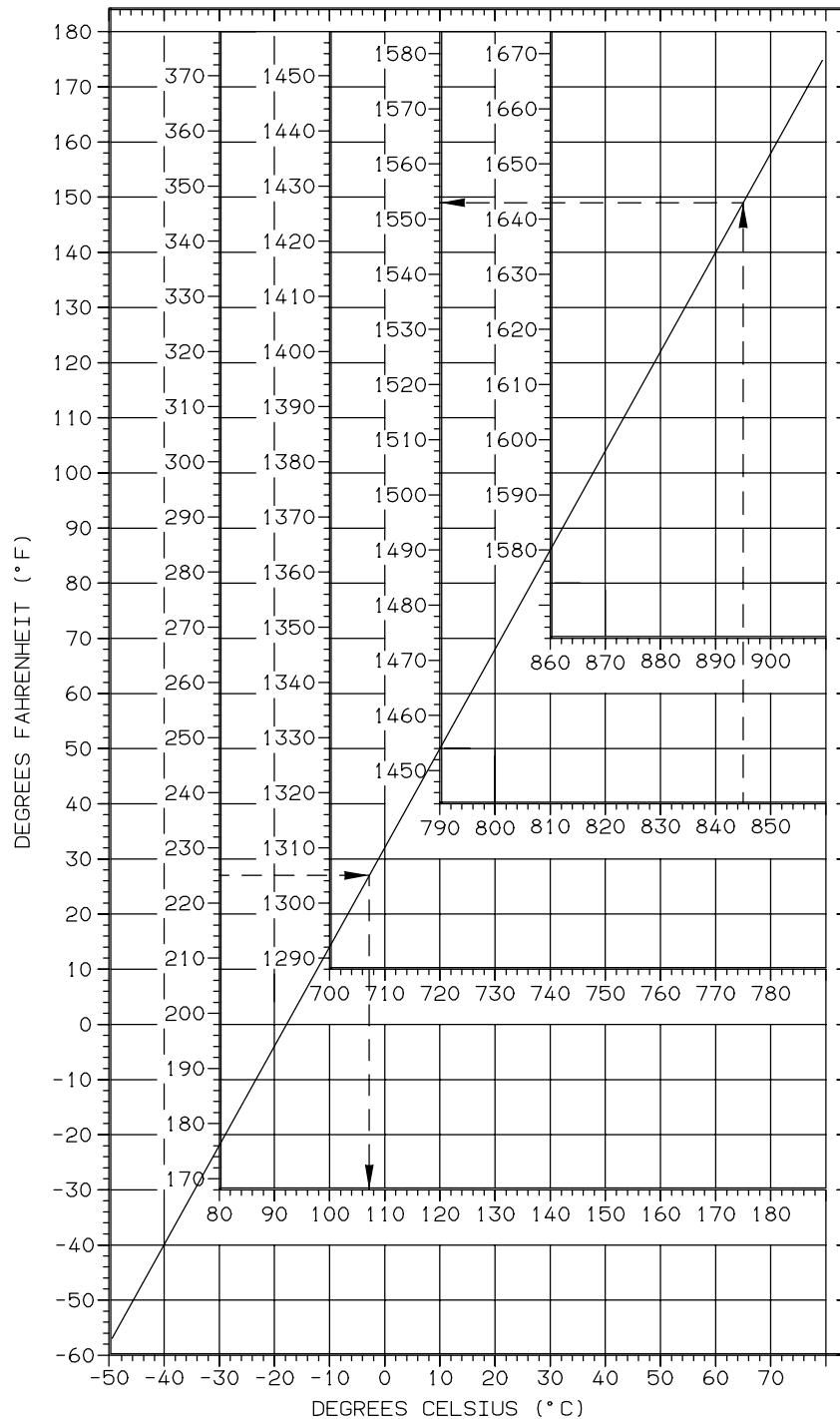
CELSIUS - FAHRENHEIT

$$^{\circ}\text{F} = 9/5 \times ^{\circ}\text{C} + 32 = 1.8 (^{\circ}\text{C} + 17.8)$$

$$^{\circ}\text{C} = 5/9 \times (^{\circ}\text{F} - 32)$$

EXAMPLES: 225 $^{\circ}\text{F}$ = 107 $^{\circ}\text{C}$

845 $^{\circ}\text{C}$ = 1553 $^{\circ}\text{F}$



BO5-1AEM,0

Fig. 1-2 Conversion Chart: Celsius - Fahrenheit

KILOMETERS PER HOUR - KNOTS

$$1 \text{ km/h} = 0.54 \text{ kt}$$

$$1 \text{ kt} = 1.853 \text{ km/h}$$

EXAMPLES: 42 kt = 78 km/h
266 km/h = 143.6 kt

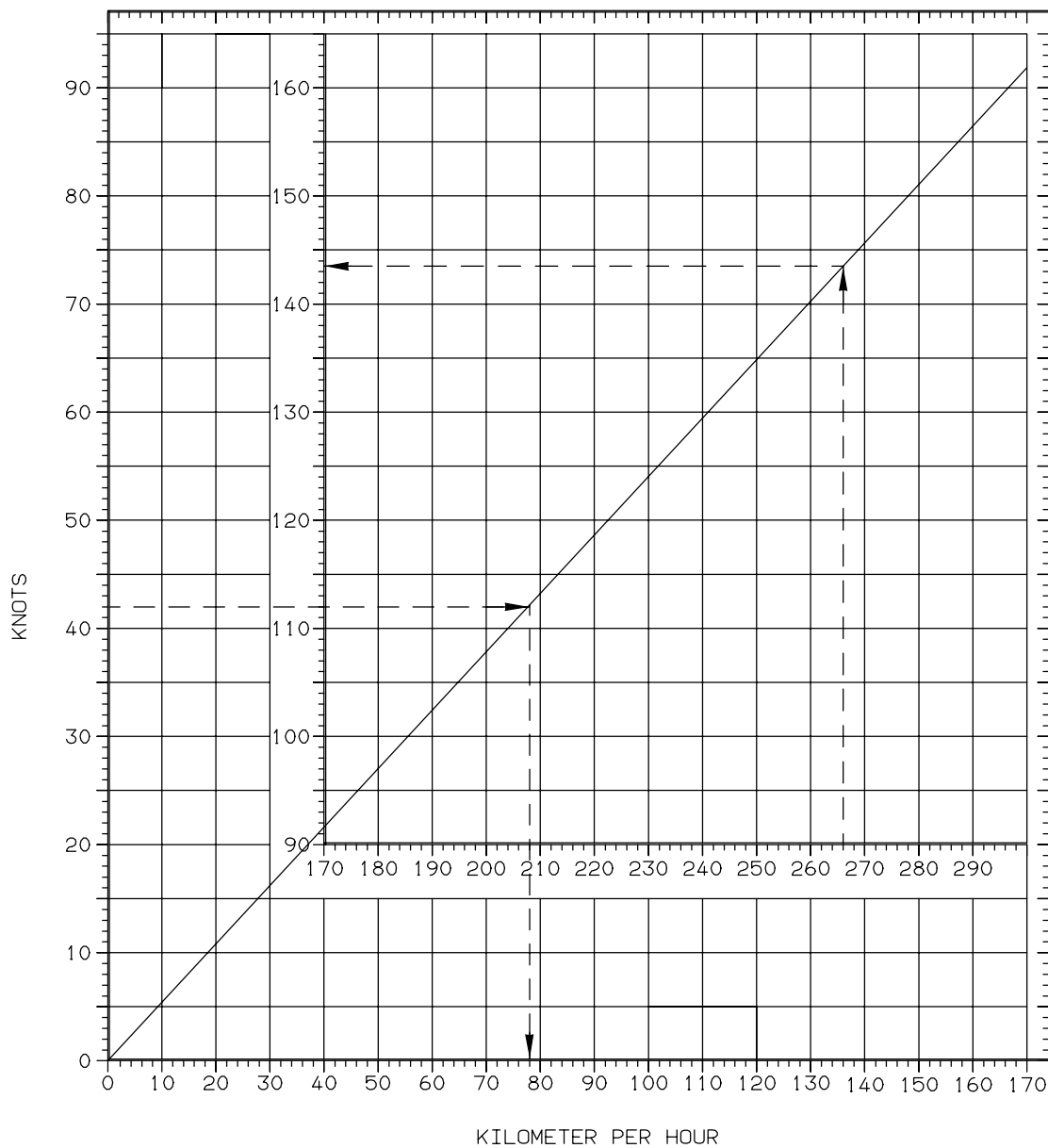


Fig. 1-3 Conversion Chart: Kilometers/Hour - Knots

METERS PER SECOND - FEET PER MINUTE

$$1 \text{ m/s} = 196.85 \text{ ft/min}$$

$$100 \text{ ft/min} = 0.508 \text{ m/s}$$

EXAMPLES: $4.7 \text{ m/s} = 925.2 \text{ ft/min}$
 $3447 \text{ ft/min} = 17.5 \text{ m/s}$

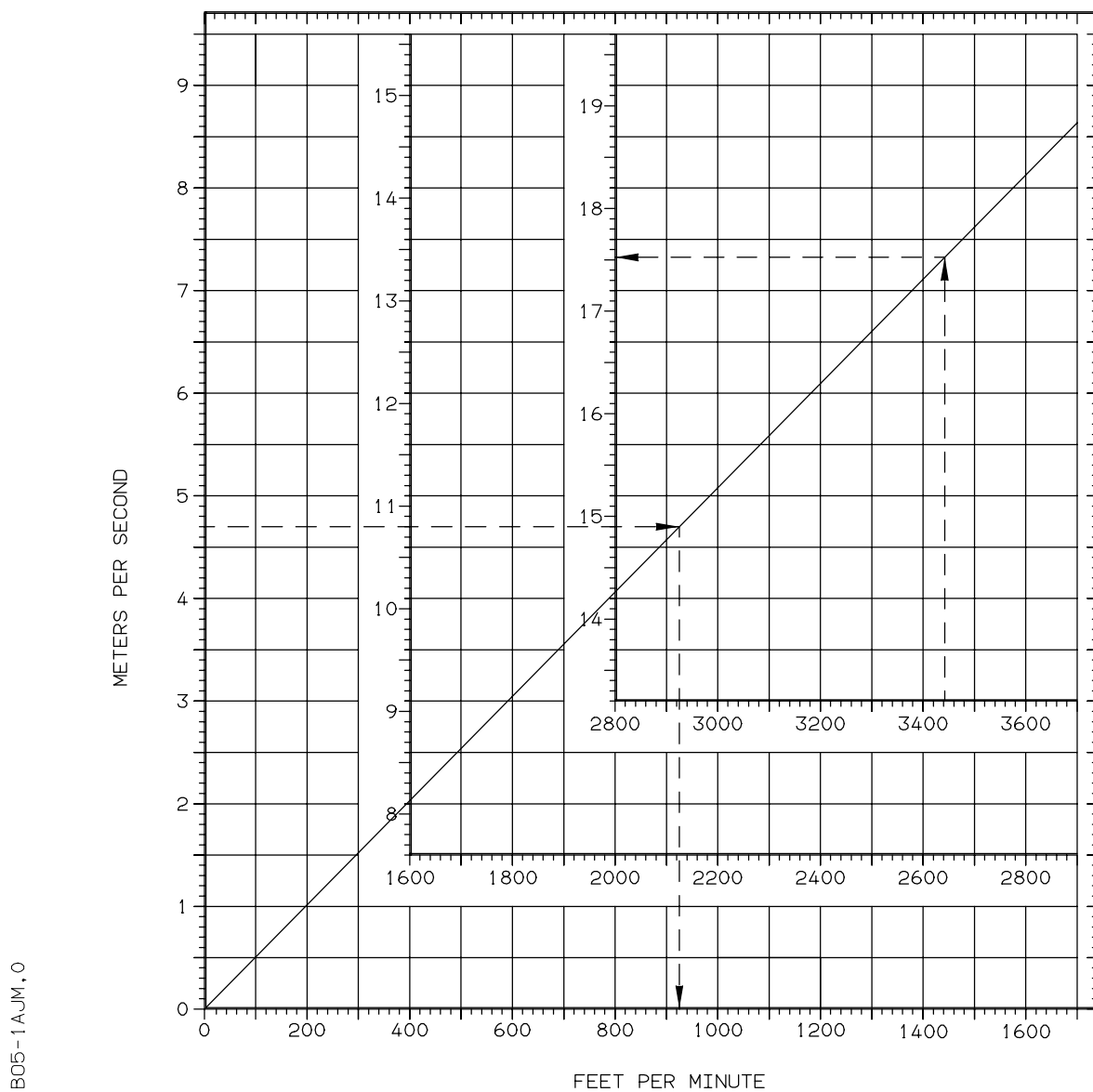


Fig. 1-4 Conversion Chart: Meters/Second - Feet/Minute

KILOGRAMS - POUNDS

$$1 \text{ kg} = 2.205 \text{ lb}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

EXAMPLES: 540 kg = 1190.7 lb
5821 lb = 2643 kg

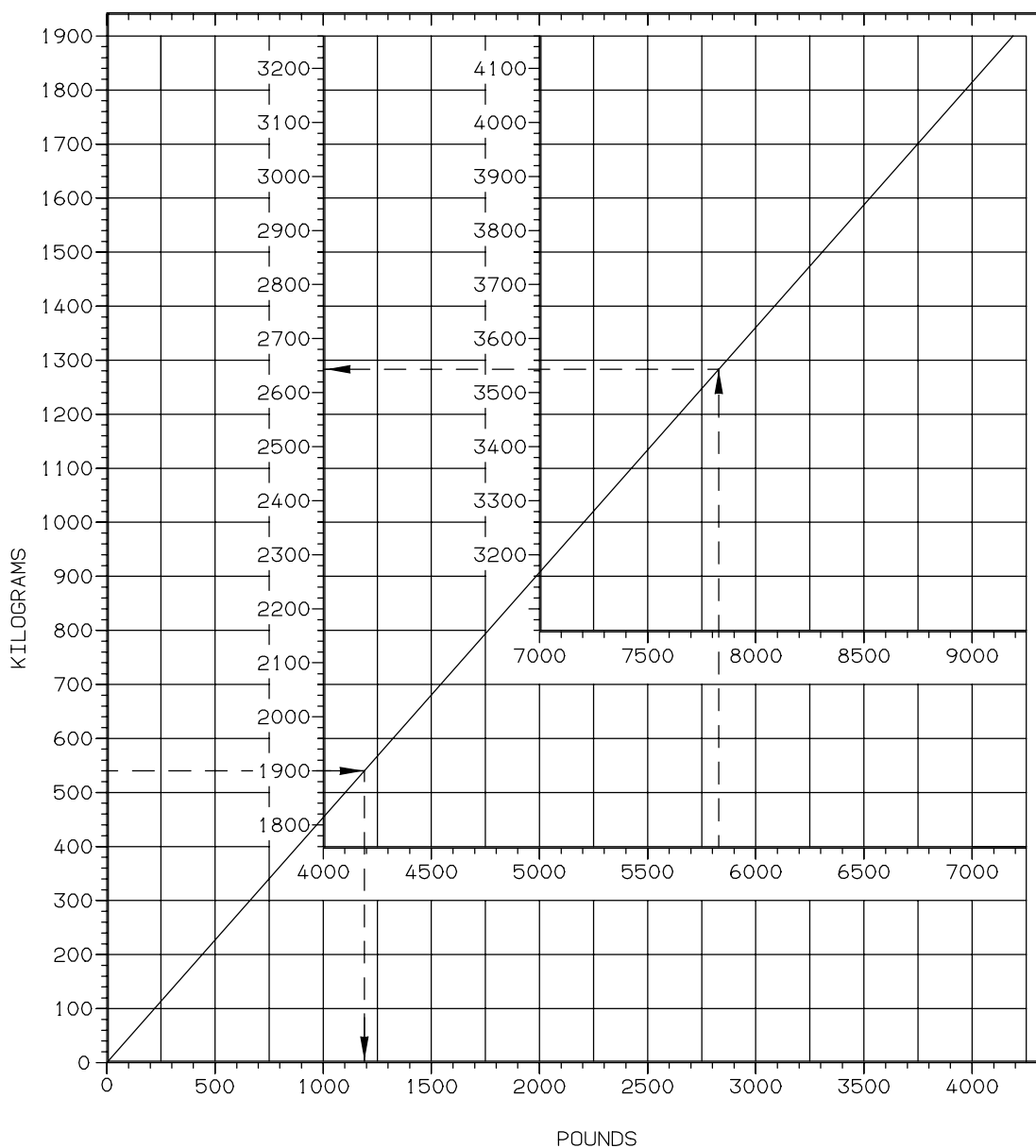


Fig. 1-5 Conversion Chart: Kilograms - Pounds

LITERS - U.S. GALLONS - IMPERIAL GALLONS



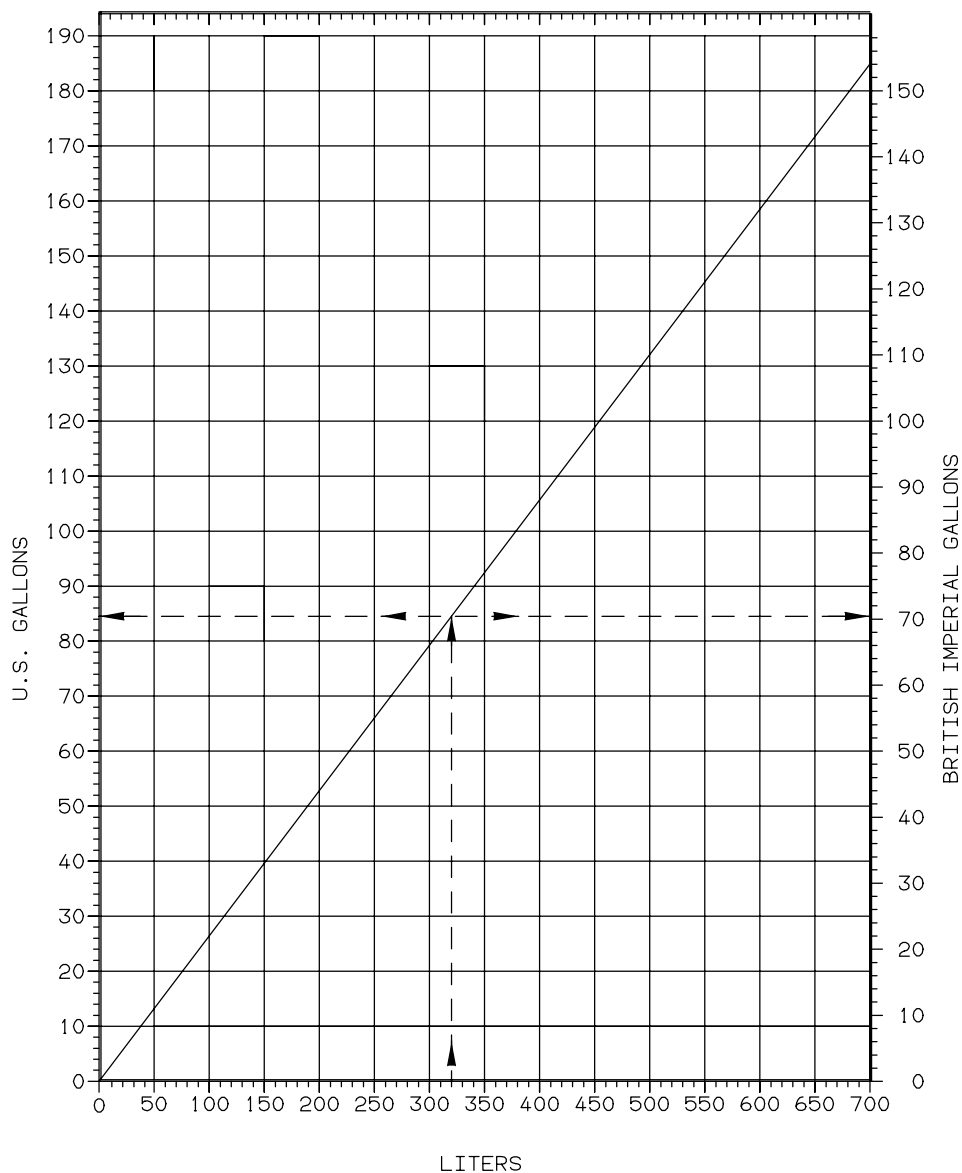
1 U.S. gal	= 3.785 l
1 l	= 0.264 U.S. gal



1 Imp. gal	= 4.546 l
1 l	= 0.220 Imp. gal

EXAMPLES:

320 l	= 84.5 U.S. gal
320 l	= 70.4 Imp. gal
70.4 Imp. gal	= 84.5 U.S. gal



B05-1ANM,0

Fig. 1-6 Conversion Chart: Liters - US Gal - Imp. Gal

HECTOPASCALS (MILLIBARS) - INCHES OF MERCURY

$$1 \text{ hPa(mbar)} = 0.02953 \text{ in. Hg}$$

$$1 \text{ in.Hg} = 33.865 \text{ hPa(mbar)}$$

EXAMPLES: $1007.5 \text{ hPa(mbar)} = 29.75 \text{ in. Hg}$
 $30.96 \text{ in. Hg} = 1048.4 \text{ hPa(mbar)}$

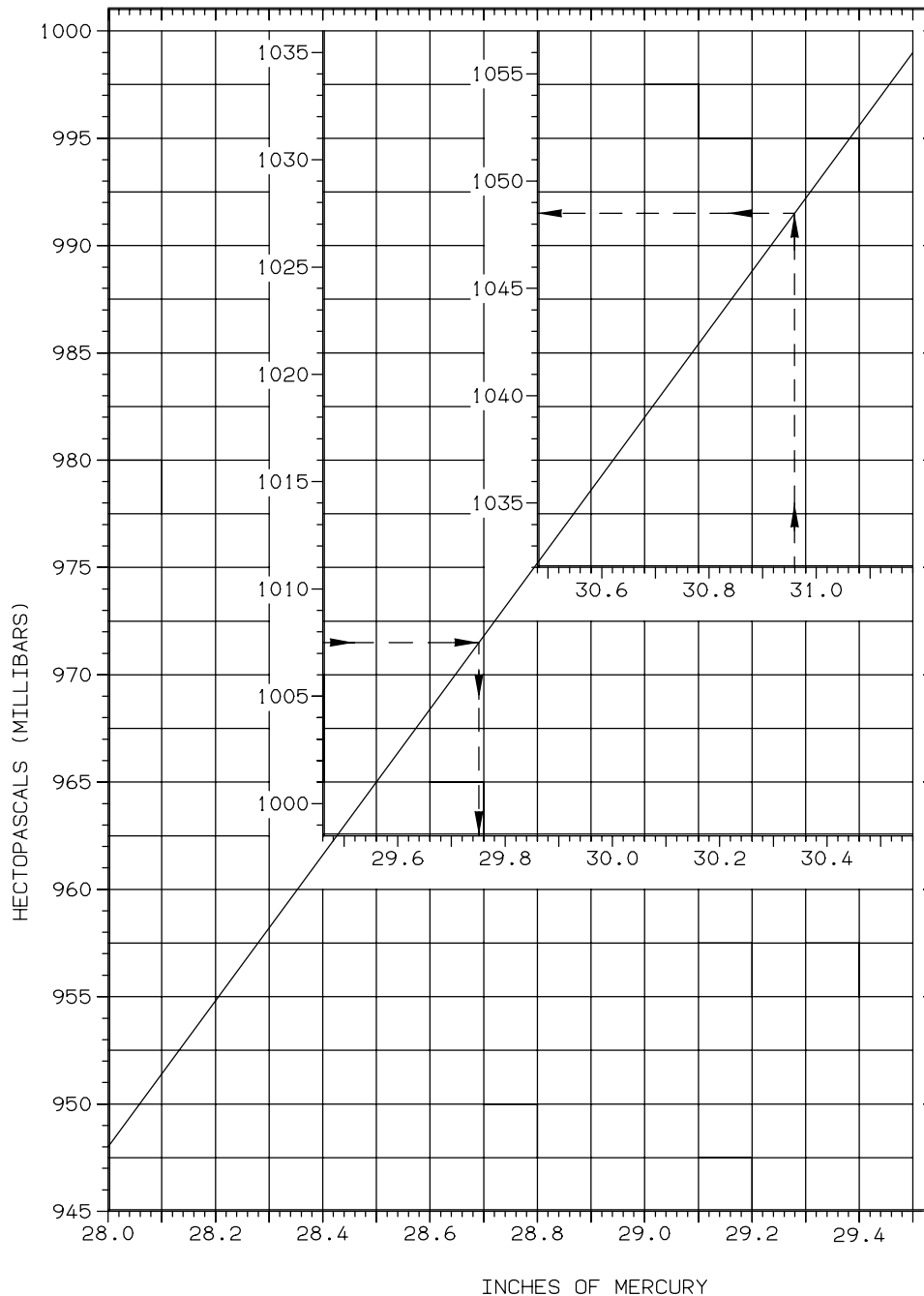


Fig. 1-7 Conversion Chart: Hectopascals - Inches of Mercury